

Case Study: Part 2: Farm Holding, EDA & Logistic regression

Your NAME

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1 The Farm holding / Desbois Case Study

1.1 Context

The article from Desbois (2008) (available [here](#) together with a sample data set) presents a complete case study around the detection of financial risks for **farm holdings** in France.

Among other methods, Linear Discriminant Analysis and Logistic Regression (plus a stepwise variable selection procedure) are used. ROC curves are provided to compare the methods.

We will reproduce and extend the Desbois case study throughout this course as a **running example** (“fil rouge”). At each stage, we apply the concepts introduced in the preceding handout.

💡 What you'll do in this Part 2

Following the concepts from [Handout 2: Logistic Regression](#):

0. Load and perform EDA on the Desbois dataset. In particular. Understand the 37 financial ratios used as predictors
1. Reuse the two scores from Part 1 (naive score based on PCA and LDA) as baselines
2. Empirical odds of Failure (+given that the farmer OWNLAND Yes/No) / Empirical odds ratio
3. Fit a Logistic Regression model with an intercept and a single qualitative predictor OWNLAND. Interpret the estimated coefficients
4. Fit a Logistic Regression model with all variables in data set, interpret the r17 coefficient
5. Compare the Scoring function/Score obtained from Logistic Regression model to the LDA and naive Score built with first component of PCA (Part 1) using a ROC curve
6. Perform statistical inference : Confidence intervals / Tests on logistic regression models (Wald, LRT) / Goodness of fit
7. Select model variables using forward stepwise procedure
8. Perform penalized logistic regression (Lasso) on the data set
9. Assess the models introduced before (naive Score, LDA, various Logistic Regression models) using resampling techniques (Hold-out and K-fold Cross Validation).

1.2 Data

1.2.1 Loading

Loading Desbois data and first glimpse at it:

```
library(tidyverse)
# package used to import spss file
library(foreign)

don_desbois <- read.spss("../data/presentation_data/Agriculture Farm Lending/desbois.sav",
glimpse(don_desbois)
```

Rows: 1,260

Columns: 30

```
$ CNTY    <fct> Eure, Eure, Eure, Eure, Eure, Eure, Eure, Eure, Eure, Eure, Eu~
$ DIFF    <fct> healthy, healthy, healthy, healthy, healthy, healthy, healthy,~
$ STATUS  <fct> Entreprise individuelle, Entreprise individuelle, Entreprise i~
```

```

$ HECTARE <dbl> 166, 101, 138, 166, 137, 107, 100, 69, 176, 177, 108, 123, 87,~
$ ToF      <fct> cereals, cereals, dairy farm, cereals, mixed livestock, cereal~
$ OWNLAND  <fct> yes, no, yes, yes, yes, yes, no, yes, yes, yes, yes, yes,~
$ AGE      <dbl> 35, 35, 42, 50, 33, 45, 34, 30, 44, 39, 40, 40, 40, 52, 49, 44~
$ HARVEST  <fct> 1988, 1988, 1988, 1988, 1988, 1988, 1988, 1988, 1988, 1988, 19~
$ r1       <dbl> 0.449, 0.450, 0.332, 0.363, 0.440, 0.306, 0.717, 0.566, 0.145,~
$ r2       <dbl> 0.622, 0.617, 0.819, 0.733, 0.650, 0.755, 0.320, 0.465, 0.881,~
$ r3       <dbl> 0.25500, 0.24110, 0.55680, 0.35960, 0.31420, 0.26350, 0.16280,~
$ r4       <dbl> 0.11450, 0.10840, 0.18510, 0.13050, 0.13820, 0.08055, 0.11680,~
$ r5       <dbl> 0.334, 0.341, 0.147, 0.232, 0.302, 0.225, 0.601, 0.499, 0.115,~
$ r6       <dbl> 0.785, 0.518, 0.700, 0.773, 0.846, 0.709, 0.894, 0.863, 0.439,~
$ r7       <dbl> 0.585, 0.393, 0.310, 0.495, 0.580, 0.523, 0.748, 0.761, 0.348,~
$ r8       <dbl> 0.20020, 0.12500, 0.38950, 0.27790, 0.26580, 0.18700, 0.14550,~
$ r11      <dbl> 0.6628, 0.7098, 0.4142, 0.4661, 0.7715, 0.8178, 0.6598, 0.6619~
$ r12      <dbl> 1.3698, 1.2534, 0.6370, 1.0698, 1.4752, 1.4682, 1.1018, 1.2360~
$ r14      <dbl> 0.23200, 0.14970, 0.48470, 0.37350, 0.25630, 0.18610, 0.18070,~
$ r17      <dbl> 0.0884, 0.0671, 0.0445, 0.0621, 0.0489, 0.0243, 0.0352, 0.0879~
$ r18      <dbl> 0.0694, 0.0348, 0.0311, 0.0480, 0.0414, 0.0173, 0.0315, 0.0758~
$ r19      <dbl> 0.1660, 0.1360, 0.1030, 0.1080, 0.1270, 0.0652, 0.1290, 0.2400~
$ r21      <dbl> 0.1340, 0.0802, 0.0890, 0.0851, 0.0838, 0.0390, 0.0784, 0.1630~
$ r22      <dbl> 0.3219, 0.3133, 0.2945, 0.1909, 0.2567, 0.1472, 0.3211, 0.5170~
$ r24      <dbl> 0.295, 0.365, 0.166, 0.265, 0.257, 0.191, 0.322, 0.305, 0.180,~
$ r28      <dbl> 0.475, 0.434, 0.350, 0.475, 0.475, 0.443, 0.401, 0.464, 0.475,~
$ r30      <dbl> 0.35000, 0.29780, 0.24680, 0.37590, 0.36690, 0.37590, 0.27240,~
$ r32      <dbl> 0.4313, 0.3989, 0.3187, 0.4313, 0.4313, 0.4257, 0.3697, 0.3886~
$ r36      <dbl> 0.886, 0.351, 1.300, 1.385, 0.886, 1.316, 0.441, 0.761, 2.147,~
$ r37      <dbl> 0.572, 0.867, 0.475, 0.470, 0.520, 0.431, 0.803, 0.656, 0.359,~

```

Replacing for convenience DIFF target by factor Y with 0 (healthy) or 1 (failing):

```

don_desbois <- don_desbois %>%
  mutate(Y = as.factor(if_else(DIFF=='healthy', 0, 1)), DIFF = NULL,
         .before = everything())

```

Your turn to play.

1.2.2 EDA

First Exploratory Data Analysis (EDA) of predictors/output:

```

##### YOUR CODE HERE #####

```

As a very soft and intuitive introduction to logistic regression. First discretize a ratio r of your choice.

- Compute proportion of defaults among each class. For example using 0.01 steps for ratio r 17:

```
##### YOUR CODE HERE #####
```

- Then using this table, plot an empirical conditional distribution of default given r classes:

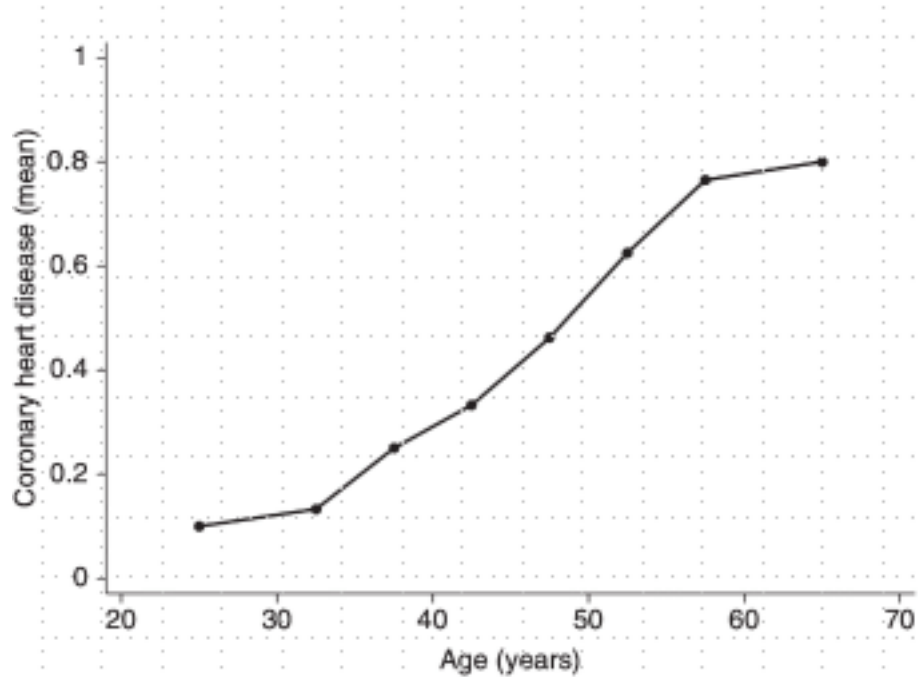
```
##### YOUR CODE HERE #####
```

This roughly corresponds to the intuitive introduction to the logistic regression model given in Hosmer et al. (2013) using a Coronary Heart Disease (CHD) event as Y and AGE as X :

The column labeled “Mean” in Table 1.2 provides an estimate of $E(Y|x)$. We assume, for purposes of exposition, that the estimated values plotted in Figure 1.2 are close enough to the true values of $E(Y|x)$ to provide a reasonable assessment of the functional relationship between CHD and AGE. With a dichotomous outcome variable, the conditional mean must be greater than or equal to zero and less than or equal to one (i.e., $0 \leq E(Y|x) \leq 1$). This can be seen in Figure 1.2. In addition, the plot shows that this mean approaches zero and one “gradually”. The change in the $E(Y|x)$ per unit change in x becomes progressively smaller as the conditional mean gets closer to zero or one. The curve is said to be *S-shaped* and resembles a plot of the cumulative distribution of a continuous random variable. Thus, it should not seem surprising that some well-known cumulative distributions have been used to provide a model for $E(Y|x)$ in the case when Y is dichotomous. The model we use is based on the logistic distribution.

Table 1.2 Frequency Table of Age Group by CHD

Age Group	n	Coronary Heart Disease		Mean
		Absent	Present	
20–29	10	9	1	0.100
30–34	15	13	2	0.133
35–39	12	9	3	0.250
40–44	15	10	5	0.333
45–49	13	7	6	0.462
50–54	8	3	5	0.625
55–59	17	4	13	0.765
60–69	10	2	8	0.800
Total	100	57	43	0.430

**Figure 1.2** Plot of the percentage of subjects with CHD in each AGE group.

1.3 Logistic Regression

1.3.1 Fitting / Coefficient interpretation

Using Farm holding data, compute (using a table for example): - The empirical odds of failure
- The empirical odds of failure given that the farmer owns his land/farm or not

Compute / interpret the empirical odds ratios (OWNLAND Yes/No)

Fit a Logistic Regression model with an intercept and a single qualitative predictor OWN-
LAND. Interpret the estimated coefficient for OWNLAND.

Fit a “full” Logistic Regression model with all predictors in data set, interpret the **r17** coefficient (in terms of probability and odds of failure):

```
##### YOUR CODE HERE #####
```

Now compare the Scoring function/Score obtained from Logistic Regression model to the LDA
and naive Score built with first component of PCA using a ROC curve:

```
##### YOUR CODE HERE #####
```

1.3.2 Confidence interval

Build a confidence interval for financial ratio **r17** and **r36** using estimated coefficients and
standard errors.

1.3.3 Tests

Considering the “full” model $Y \sim .$, assess the statistical significance of **r36**? Using Wald test.
Using Likelihood Ratio Test.

1.3.3.1 Wald Test

```
# YOUR CODE HERE
```

1.3.3.2 LRT Test

```
# YOUR CODE HERE
```

Fit a “large” model with 9/10 financial ratios (including `r17` and `r36`) as predictors. Fit a “smaller” model removing `r17` and 2/3 additional ratios of your choice. Compare using LRT the two model fit, which one would you retain?

1.3.3.3 Goodness of fit tests

Perform a goodness of fit test (Hosmer and Lemeshow) and a calibration plot, compare.

```
# YOUR CODE HERE
```

1.4 Model selection and assessment

1.4.1 Stepwise Logistic Regression

Use forward stepwise selection based on the AIC criterion to select variables in the Farm holding data set. Compare with results from the article. Do you find the same variables as the author? (Extra, perform the same analysis using LDA/Wilks Lambda forward selection procedure as in the article).

```
# YOUR CODE HERE
```

1.4.2 Penalized Logistic Regression

Perform penalized logistic regression (Lasso) on the Farm holding data set. In this context, penalization is used as an automatic variable selection tool: it will shrink the coefficients of some predictors exactly to zero. Select a best penalization parameter using K-fold cross validation. You can find help [here](#).

```
# YOUR CODE HERE
```

1.4.3 Model assessment

1.4.3.1 Hold-out

Perform a simple Hold-out approach (train/test split) and evaluate ROC / AUC for some `glm()` model specification of your choice (ie. choose manually a subset of variables), you

might take inspiration from this [book chapter](#). You can also compare the former Scoring functions already used in the case study (LDA, Naive score, ...).

```
# YOUR CODE HERE
```

1.4.3.2 K-fold cross validation

Perform k-fold cross validation and for each fold evaluate ROC / AUC for the same models as before.

```
# YOUR CODE HERE
```

Desbois, D. (2008). Introduction to scoring methods: financial problems of farm holdings. *Case Studies in Business, Industry and Government Statistics*, 2(1), 56–76. <https://hal.science/hal-01172847>

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). Applied logistic regression. In *Wiley Series in Probability and Statistics*.